



GEETHANJALI INSTITUTE OF SCIENCE & TECHNOLOGY

(AN AUTONOMOUS INSTITUTION)

(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu)

(Accredited by NAAC with "A" Grade, NBA (EEE, ECE & ME) & ISO 9001:2008 Certified Institution)

QUESTION BANK (DESCRIPTIVE)

Subject Name: Block Chain Technology

Subject Code: 23A0538Tb

Course & Branch: B.Tech C S E (C S)

Year & Semester: IV B.Tech I Semester

Regulation: RG23

Unit 1

S. No	2 Marks Questions (Short)	[BT Level] [CO] [Marks]
1.	Define Blockchain.	L1, CO1, 2M
2.	List any four characteristics of Blockchain.	L1, CO1, 2M
3.	What is a centralized application? Give an example.	L1, CO1, 2M
4.	Define "Fork" in Blockchain.	L1, CO1, 2M
5.	What is a Consortium (federated) blockchain?	L1, CO1, 2M
6.	Mention any two opportunities offered by Blockchain technology.	L1, CO1, 2M
7	Differentiate Public and Private Blockchain (any two points).	L1, CO2, 2M
8.	When and by whom was Bitcoin introduced?	L1, CO2, 2M
9	Name any two types of players in the Blockchain ecosystem.	L1, CO2, 2M
10	What is a decentralized application (DApp)? Give an example.	L1, CO2, 2M

S. No	Descriptive Questions (Long)	[BT Level] [CO] [Marks]
1	Explain the concept of Blockchain and its key characteristics in detail.	L2, CO1, 10M
2	Discuss real-world scenarios that illustrate business/technical challenges leading to Blockchain adoption.	L2, CO1, 10M
3	Explain the opportunities Blockchain offers across industries, with examples.	L2, CO1, 10M
4	Trace the history and evolution of Blockchain technology with major milestones.	L2, CO1, 10M
5	Explain the evolution of computer applications from centralized to decentralized systems.	L2, CO1, 10M
6	Explain the stages in Blockchain evolution (Blockchain 1.0, 2.0, 3.0) with use cases.	L2, CO2, 10M
7	What is a Consortium blockchain? Explain its features, working, and advantages over public/private blockchain	L2, CO2, 10M

8	Compare Centralized vs Decentralized Applications (architecture, control, security, scalability).in detail?	L2,CO2,10M
9	Explain Forks in Blockchain — types (hard fork, soft fork) with examples, and Public Blockchain environments and their features.	L2,CO2,10M
10	Explain the types of players in the Blockchain ecosystem, and describe key players currently active in the market.	L2,CO2,10M
Unit-2		
S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	Define a "block" in Blockchain and mention what happens when a block is tampered with.	L1,CO3,2M
2.	What is Hashing? Name one hashing algorithm used in Blockchain.	L1,CO3,2M
3.	Define Merkle Tree. Why is it used in Blockchain?	L1,CO3,2M
4.	What is Consensus in Blockchain?	L1,CO3,2M
5.	Define Mining in Blockchain.	L1,CO3,2M
6.	What are tokens (cryptocurrency) in Blockchain?	L1,CO3,2M
7.	What is a Blockchain wallet? Name its two types.	L1,CO3,2M
8.	Define Smart Contract	L1,CO3,2M
9	What is a Peer-to-Peer (P2P) network?	L1,CO3,2M
10	Name any two types of blockchain nodes.	L1,CO3,2M
S.No	Descriptive Questions(Long)	[BT Level] [CO][Marks]
1	Explain the basic concept of Blockchain, and describe how a block is changed/added and how tampering affects the chain's integrity.	L2,CO3,10M
2	Explain Hashing and its role in Blockchain, and describe the structure and significance of the Merkle Tree in data verification.	L2,CO3,10M
3	Explain Consensus in Blockchain. Describe any two consensus mechanisms	L2,CO3,10M
4	Explain the process of Mining and how blocks are validated and finalized on the Blockchain.	L3,CO3,10M
5	Discuss currency/tokens in Blockchain — coins vs tokens, and their role in the blockchain economy.	L2,CO3,10M
6	Explain the security mechanisms used in Blockchain to protect data, transactions, and the network from attacks	L2,CO3,10M
7	Explain how data is stored on a Blockchain, and describe the types and working of Blockchain wallets.	L2,CO3,10M
	What are Smart Contracts? Explain coding on Blockchain features.	L2,CO3,10M

8	advantages, and a real-world use case.	
9	Explain the Peer-to-Peer network architecture of Blockchain and describe the different types of blockchain nodes and their functions.	L2,CO3,10M
10	Explain the life cycle of a blockchain transaction from initiation to confirmation, and discuss the key risks associated with blockchain solutions.	L2,CO3,10M

UNIT-3

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	What is meant by "Architecting a Blockchain Solution"?	L1,CO4,2M
2.	List any two obstacles/challenges in adopting Blockchain for enterprise use.	L1,CO4,2M
3.	What is a Blockchain Relevance Evaluation Framework?	L1,CO4,2M
4.	Define Reference Architecture in the context of Blockchain solutions.	L1,CO4,2M
5.	Name any two types of Blockchain applications.	L1,CO4,2M
6.	What is a Cryptographic Token?	L1,CO4,2M
7.	What are the different types of Blockchain solutions ?	L1,CO4,2M
8.	Name any two architecture considerations while designing a Blockchain solution.	L1,CO4,2M
9	Name any two Blockchain platforms used for building enterprise architecture.	L1,CO4,2M
10	What is the first step in the approach to designing a Blockchain application?	L1,CO4,2M

S. No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Explain the concept of Architecting Blockchain Solutions and its importance for enterprises.	L2,CO4,10M
2	Discuss the key obstacles faced in the adoption and use of Blockchain technology, with examples.	L2,CO4,10M
3	Explain the Blockchain Relevance Evaluation Framework — how it helps determine whether Blockchain is suitable for a given problem.	L2,CO4,10M
4	Describe the Blockchain Solutions Reference Architecture with its components and layers.	L2,CO4,10M
5	Explain the different Types of Blockchain Applications with real-world examples.	L2,CO4,10M
6	What are Cryptographic Tokens? Explain their types (utility, security,	L2,CO4,10M

	payment tokens) and role in Blockchain solutions.	
7	Describe a Typical Solution Architecture for Enterprise Use Cases of Blockchain, with a suitable diagram/example.	L2,CO4,10M
8	Explain the various Types of Blockchain Solutions (e.g., record-keeping, asset transfer, identity, smart contracts) with examples.	L2,CO4,10M
9	Discuss the key Architecture Considerations while designing a Blockchain solution, and explain how architecture varies across different Blockchain Platforms	L4,CO4,10M
10	Explain the step-by-step Approach for Designing Blockchain Applications, from problem identification to deployment.	L2,CO4,10M

Unit-4

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	What is Ethereum? Mention its role in Blockchain.	L1,CO5,,2M
2.	What is the Tuna Fish Tracking use case in Blockchain?	L1,CO5,2M
3.	Define Ethereum Virtual Machine (EVM).	L1,CO5,2M
4.	What is Solidity?	L1,CO5,2M
5.	What is the Truffle Framework used for?	L1,CO5,2M
6.	What is Ganache? Mention its use in development.	L1,CO5,2M
7.	Define an Ethereum Account and name its two types.	L1,CO5,2M
8.	What is MetaMask?	L1,CO5,2M
9	What is Infura, and why is it used?	L1,CO5,2M
10	Define Decentralized Application (DApp).	L1,CO5,2M

S.No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Explain the Ethereum Ecosystem and its key components?in detail?	L2,CO5,10M
2	Describe the Tuna Fish Tracking use case — the business problem, and how Blockchain addresses it.	L2,CO5,10M
3	Explain the Ethereum Development process and describe the Ethereum Tool Stack used by developers.	L2,CO5,10M
4	Explain the architecture and working of the Ethereum Virtual Machine (EVM)	L2,CO5,10M
5	Explain Smart Contract Programming — language used (Solidity), structure, compilation, and deployment.	L2,CO5,10M
6	Discuss the Integrated Development Environment (IDE) options for Ethereum, and explain the Truffle Framework with its features	L2,CO5,10M
7	Explain Ganache and its role in local blockchain development, and describe Unit Testing of smart contracts.	L4,CO5,10M
8	Explain Ethereum Accounts (types and structure), and describe wallets — My Ether Wallet and MetaMask — with their features.	L2,CO5,10M
9	Explain Ethereum Networks/Environments (Mainnet, Testnets), Ethereum Clients, and the roles of Infura and Etherscan as supporting infrastructure/tools.	L2,CO5,10M
10	Explain Decentralized Application (DApp) development, and describe the implementation of the Tuna Fish Use Case using OpenZeppelin contracts.	L2,CO5,10M

Unit-5

S.No	2MarksQuestions(Short)	[BTLevel][CO][Marks]
1	What is Hyperledger Fabric? Mention one feature that distinguishes it from public blockchains.	L1,CO6,2M
2	define Car Ownership Tracking use case.	L1,CO6,2M
3	What is a Chaincode in Hyperledger Fabric?	L1,CO6,2M
4	Name the key components involved in the Hyperledger Fabric transaction flow.	L1,CO6,2M
5	What is IPFS (InterPlanetary File System)?	L1,CO6,2M
6	Define Zero Knowledge Proof (ZKP)	L1,CO6,2M
7	What is an Oracle in Blockchain?	L1,CO6,2M
8	Define Self-Sovereign Identity (SSI).	L1,CO6,2M
9	What is an Initial Coin Offering (ICO)?	L1,CO6,2M
10	Name any two Blockchain Cloud Offerings	L1,CO6,2M

S. No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Explain the introduction to Hyperledger Blockchain and describe the Car Ownership Tracking use case in detail.	L4,C06,10M
2	Explain Hyperledger Fabric its architecture and key components?In Detail?	L4,C06,10M
3	Describe the Hyperledger Fabric Transaction Flow step by step, from proposal to commitment.	L2,C06,10M
4	Explain the implementation of the Fabric-based use case (Car Ownership Tracking) with its design and workflow.	L2,C06,10M
5	Explain how Chaincode functions are invoked using a Client Application, with an example.	L2,C06,10M
6	Explain the InterPlanetary File System (IPFS) and its role in decentralized storage integration with Blockchain.	L2,C06,10M
7	Explain Zero Knowledge Proofs and Oracles, and describe their significance in enhancing Blockchain privacy and connecting to external data.	L4,C06,10M
8	Explain Self-Sovereign Identity (SSI) and discuss the integration of Blockchain with IoT and AI/ML.	L2,C06,10M
9	Discuss Quantum Computing's impact on Blockchain security, and explain the concept of Initial Coin Offerings (ICOs)	L2,C06,10M
10	Explain Blockchain Cloud Offerings by major providers, and discuss the	L2,C06,10M

	future potential of Blockchain technology.	
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